

WOXSEN UNIVERSITY

SCHOOL OF TECHNOLOGY • DEPARTMENT OF AI & ML

IT HELPDESK & ASSET SUPPORT MANAGEMENT SYSTEM

Course Project Report & System Demonstration

Course: Database Management Systems (DBMS)

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Section: AIML Panthers

Phase: Presentation-III & Final Project Evaluation

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1. Abstract

The **IT Helpdesk and Asset Support Management System** is a comprehensive relational database application engineered for enterprise technical support and computing hardware asset lifecycle tracking. Modern organizations frequently suffer from lost helpdesk requests, missing computing assets, unmonitored repair expenses, and absent operational audit trails due to fragmented management tools.

This project delivers a normalized **Third Normal Form (3NF)** relational database designed in **MySQL 8.0+** comprising 14 interconnected tables. The database enforces relational integrity using Primary Keys, Foreign Keys (`CASCADE`, `RESTRICT`, `SET NULL`), `UNIQUE` constraints, and `CHECK` expressions. The system features a modern Python backend using parameterized SQL queries and a single-page web interface. Real-time **VIEW**, **INSERT**, and **DELETE** operations are supported with live verification in MySQL. Automated testing confirms 100% compliance across all evaluation criteria.

2. Introduction & Problem Statement

Modern enterprises manage diverse computing assets alongside dozens of daily technical support requests across multiple organizational departments. When managed through spreadsheets or informal channels:

- **Lost Helpdesk Tickets:** Tickets lack centralized queuing, causing SLA breaches and neglected users.
- **Technician Accountability Deficit:** Unclear technician assignments result in duplicate effort and delayed resolutions.
- **Untracked Hardware Inventory:** Relocations, warranty expirations, and device reallocations go unmonitored.
- **Absence of Immutable Audit Trails:** State transitions are overwritten rather than historically preserved.
- **Untracked Maintenance Expenditure:** Servicing expenses are not correlated with physical assets or departments.

3. System Objectives & Scope

1. Provide a centralized MySQL relational database in Third Normal Form (3NF).
2. Support entity inheritance for **Incidents** (failures) and **Service Requests** (provisioning).
3. Catalog hardware assets by unique tag and serial number, linked to user ownership and warranties.
4. Record immutable audit trails in `status_histories` for every state transition.
5. Deliver a live, responsive web user interface connected directly to MySQL for live viva demonstration.

4. Relational Schema & 3NF Normalization

The database schema comprises 14 normalized tables:

- **Organization Domain:** departments, users
- **Support Classification:** categories, priorities
- **Support Staff:** support_staff, assignments
- **Hardware Assets:** warranties, assets, maintenance
- **Ticketing & Lifecycles:** tickets, incidents, service_requests, status_histories, resolutions

Normalization Proof:

- **1NF:** All attributes contain atomic values; composite/multi-valued attributes are separated into dedicated relations.
- **2NF:** Every non-prime attribute is fully functionally dependent on its respective primary key.
- **3NF:** No non-prime attribute is transitively dependent on a primary key (e.g. departmental locations depend on department_id, not user_id).

5. Testing & Verification Results

Test ID	Test Description	Expected Result	Actual Result	Status
TC01	MySQL Database Connection	Successful connection, 14 tables verified	Connected to it_helpdesk (v9.7.1, 14 tables)	PASS
TC02	View Tickets	Relational rows retrieved via JOINS	5 tickets loaded with user & category info	PASS
TC03	Insert Ticket Record	Atomic multi-table insertion	Created TKT-006 with subtype & history log	PASS
TC04	Refresh & Persistence	Record confirmed in MySQL InnoDB storage	Row verified directly via SQL query	PASS
TC05	Delete Ticket	Record removed from MySQL	Cascade deleted from tickets and child tables	PASS
TC06	Absence After Delete	Record remains absent after refresh	Verified 0 rows in MySQL for deleted ID	PASS
TC07	Form Input Validation	HTTP 400 rejection on empty required fields	Rejected with friendly validation message	PASS
TC08	Unique Constraint	Rejection on duplicate user email	Caught duplicate email with HTTP 409	PASS
TC09	Foreign Key Deletion	Friendly warning on deleting user with tickets	Caught FK RESTRICT violation with friendly error	PASS
TC10	MySQL Offline Resilience	Safe user error without leaking credentials	Displays clean database offline message	PASS

6. Conclusion & Viva Demonstration Summary

The **IT Helpdesk and Asset Support Management System** fulfills all official academic criteria for Presentation-III:

1. Connected directly to the real local MySQL server on port 3306.
2. Supports interactive **VIEW**, **INSERT**, and **DELETE** operations.
3. Enforces referential integrity with cascading deletions and protective restrictions.
4. Provides live database persistence verified across browser refreshes and direct SQL checks.

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